LOAN PREDICTION USING MACHINE LEARNING

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INTRODUCTION

Loan prediction is an important application of Machine Learning in the banking and finance sector. It helps financial institutions determine whether a loan applicant is eligible for a loan based on their personal and financial information.

Banks receive many loan applications every day. Manually verifying each application is time-consuming and may lead to errors. Machine Learning models analyze historical loan data and predict whether a loan should be approved or rejected.

This project aims to develop a Machine Learning model that predicts loan approval using applicant details such as income, education, employment status, loan amount, credit history, and other financial information.

The developed system helps banks reduce risk, improve decision-making, and speed up the loan approval process.

PROBLEM STATEMENT

Banks face the challenge of approving loans while minimizing financial risk. Approving loans for applicants who may fail to repay can result in significant losses.

By analyzing historical loan data, Machine Learning models can identify patterns that indicate whether an applicant is likely to repay the loan. Loan prediction enables banks to make faster, more accurate, and data-driven lending decisions.

OBJECTIVES

The primary objectives of this project are:

* To analyze applicant demographic and financial information.
* To preprocess and clean the loan dataset.
* To perform Exploratory Data Analysis (EDA).
* To develop Machine Learning models for loan prediction.
* To predict whether a loan application should be approved or rejected.
* To support banks in making accurate loan approval decisions.

TECHNOLOGIES USED

The following technologies and tools were used for the development of the Loan Prediction System:

1. Python
    * Programming language used for model development.
2. Google Colab
    * Cloud-based platform used for coding and execution.
3. Pandas
    * Used for data manipulation and analysis.
4. NumPy
    * Used for numerical computations.
5. Matplotlib
    * Used for creating data visualizations.
6. Seaborn
    * Used for statistical data visualization.
7. Scikit-learn
    * Used for Machine Learning model development and evaluation.

Machine Learning Algorithms Used

* Logistic Regression
* Decision Tree Classifier
* Random Forest Classifier

DATASET DESCRIPTION

* Dataset Name: Loan Prediction Dataset
* Source: Kaggle
* Total Records: 614
* Total Features: 13
* Target Variable: Loan_Status (Approved/Rejected)

Dataset Features

* Loan_ID
* Gender
* Married
* Dependents
* Education
* Self_Employed
* ApplicantIncome
* CoapplicantIncome
* LoanAmount
* Loan_Amount_Term
* Credit_History
* Property_Area
* Loan_Status